

YASH KARTHIYA

B.Tech Student | RAG, LLM Evaluation & Backend Systems

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Computer Engineering student building backend-heavy GenAI systems, with project experience in RAG pipelines, LLM evaluation, and FastAPI microservices focused on retrieval optimization and latency benchmarking.

TECHNICAL EXPERTISE

LLM Systems	RAG, LLM Evaluation, Retrieval Benchmarking, Prompt Design, Re-ranking
Backend	FastAPI, Node.js, Spring Boot, Java 17, Python, Asynchronous & Event-Driven Pipelines
Data / Search	ChromaDB, Vector Embeddings, PostgreSQL, MongoDB, Redis, Semantic Search
Infrastructure	Docker, AWS (EC2, S3), CI/CD, Linux

SELECTED PROJECTS

Academic Query Engine with Evaluation-Driven RAG

Dec 2025 – Mar 2026

FastAPI, Python, ChromaDB, LangChain, OpenAI

[View on GitHub](#)

- Built a context-aware RAG pipeline over 500+ documents, achieving **sub-500ms p95 retrieval latency** in **Locust** benchmarks.
- Improved retrieval relevance by 30%** versus a baseline cosine-search pipeline, measured on 1,000 internally created benchmark queries.
- Designed a custom evaluation framework to measure **retrieval accuracy, answer relevance, and hallucination rate** across varying retrieval depths.
- Improved grounding by adding cross-encoder re-ranking and prompt filtering to produce more **evidence-aligned answers**.

SafePlate: Multimodal Food Scanner — DUHacks 5.0 (2026)

FastAPI, React, Tesseract OCR, OpenAI

[View on GitHub](#)

- Engineered an OCR-to-LLM pipeline in a 36-hour sprint to convert unstructured packaging data into **structured nutrition and safety insights**.
- Integrated LLM-based reasoning to classify ingredients and compute safety scores, surfaced through a real-time React dashboard.
- Evaluated OCR preprocessing across **100+ product images** to improve extraction quality on noisy and diverse real-world packaging layouts.

SkillSwap: Real-Time P2P Learning Platform

Jul 2025 – Nov 2025

Node.js, Express, React, MongoDB, Socket.io, WebRTC

[View on GitHub](#)

- Built a P2P signaling system handling **50+ concurrent WebRTC sessions** with sub-100ms connection setup for interactive collaboration.
- Engineered Redis-based state synchronization to manage multi-user sessions, reducing stale connection overhead by 25% in internal load tests.

High-Concurrency Shop Management System

Dec 2024 – Mar 2025

Spring Boot, PostgreSQL, Redis, Java 17, Docker

[View on GitHub](#)

- Optimized transaction latency to **sub-50ms for SKU lookups** by implementing write-through Redis caching for hot SKUs.
- Used Spring Events for asynchronous state updates, decoupling database writes from the user request cycle to improve system throughput.

EDUCATION

Bachelor of Technology in Computer Engineering

2023 – 2027

Dharmsinh Desai University | CGPA: 8.29 / 10.0

Nadiad, Gujarat

ADDITIONAL INFORMATION

- Project Demo & Sprint:** Built and demoed an end-to-end vision AI system during a 36-hour sprint at DUHacks 5.0.
- Deployment:** Deployed project demos and portfolio services on AWS for peer testing and live demonstration.
- Languages:** English (Professional), Hindi (Native), Gujarati (Native).